Last time:

The matrix M defines an inner product on fin dim vector space V $\langle x, y\rangle = x^T M y$ for some symmetric matrix w/ strictly positive eigenvalues

EX find an inner product on $\mathbb{R}^2$ s.t. $\begin{bmatrix}1\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\end{bmatrix}$ are orthogonal, $\theta = \pi/2$ looking for M s.t. $0 = [1\ 0] M \begin{bmatrix}1\\1\end{bmatrix}$

$$0 = [1\ 0]\underbrace{\begin{bmatrix}a & b\\ c & d\end{bmatrix}}_{b=c}\begin{bmatrix}1\\1\end{bmatrix} = [1\ 0]\begin{bmatrix}a+b\\ b+d\end{bmatrix} = 0 = (a+b) + 0(b+d) \Rightarrow a+b=0$$

$$a = -b$$

$$M = \begin{bmatrix}a & -a\\ -a & d\end{bmatrix} \qquad \begin{vmatrix}a-\lambda & -a\\ -a & d-\lambda\end{vmatrix} = 0 \quad (a-\lambda)(d-\lambda) - a^2 = 0$$

$$\lambda^2 - (a+d)\lambda + ad - a^2 = 0$$

$$\lambda = \frac{a+d \pm \sqrt{(a+d)^2 - 4(ad-a^2)}}{2} \qquad a+d - \sqrt{(a+d)^2 - 4(ad-a^2)} > 0$$

$$a+d > \sqrt{(a+d)^2 - 4(ad-a^2)}$$

$$a+d > \sqrt{a^2+d^2+2ad-4ad+4a^2} \qquad a+d > \sqrt{5a^2-2ad+d^2} \qquad 5a^2+d^2 \geq 2ad$$

try $a=d=1$ $\quad 5+1 \geq 2$ ✓ $\quad$ try $a=1$ $d=2$ $\quad 9 \geq 4$, $3 > \sqrt{5}$ ✓

$2 > 2$ ✗

$$\langle x, y\rangle = x^T \begin{bmatrix}1 & -1\\ -1 & 2\end{bmatrix} y \quad \text{check for } x = \begin{bmatrix}1\\0\end{bmatrix}\ y = \begin{bmatrix}1\\1\end{bmatrix}$$

$$\left\langle \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\rangle = 0 = [1\ 0]\begin{bmatrix}1 & -1\\ -1 & 2\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = [1\ 0]\begin{bmatrix}0\\1\end{bmatrix} = 0 \checkmark$$

---

intuition: bases should be chosen to make the problem at hand easy and useful, inner products are the same way